

Jackson Brouwer

607-592-9338 | jbrouwer08@gmail.com | linkedin.com/in/jackson-brouwer | github.com/jbrouwer5

EDUCATION

University of Chicago

Chicago, IL

Master of Science in Computer Science

Dec. 2023 – Aug. 2024

- GPA: 3.81/4.00
- Featured Coursework: Advanced C++, Databases, HPC, NLP, Parallel Programming, Time-Series Analysis

University of Chicago

Chicago, IL

Bachelor of Science in Computer Science

Sept. 2020 – Dec. 2023

- GPA: 3.70/4.00, Dean's List: 2021-2022
- Featured Coursework: Algorithms, Architecture, Computer Vision, Data Structures, Mobile Computing, Systems

EXPERIENCE

Software Engineering Intern

June 2023 – Jan. 2024

Zelia (Techstars NYC '23)

New York, NY

- Headed development and maintenance of RESTful APIs for core platform features, utilizing **Python** and SQLAlchemy to support client-facing apps with over 75,000 users
- Architected and deployed scalable AWS infrastructure, enhancing system reliability, availability, and performance
- Built CI/CD pipelines with PyTest and GitHub Actions, automating transactional queries for efficient deployment
- Led migration of on-site PostgreSQL database to AWS for 50,000-user dataset, optimizing with column indexing and composite primary keys, achieving a 30% reduction in average query latency

Software Engineering Intern

June 2022 – Sept. 2022

Fuel Cycle

Los Angeles, CA

- Developed back-end QA tests for an enterprise market insights automation platform
- Built automated unit and functional API tests in Java and Postman, identifying 22% more bugs before production
- Collaborated with front-end engineers to resolve issues involving back-end API integration

Undergraduate Research Assistant

Apr. 2021 – Feb. 2022

University of Chicago

Chicago, IL

- Assessed efficacy of smart-home IoT devices through simulation experiments and guided user sessions
- Developed and iterated upon full-stack IoT simulation software with Django and Angular

Non-Profit Lead Developer

Sept. 2021 – Jan. 2022

University of Chicago IOP

Chicago, IL

- Led four developers to implement iOS compatibility using React in an educational app for Chicago Public Schools
- Resulted in 60% higher monthly user count from new iOS users

PROJECTS

Handwriting in VR | *Python, C#, AWS*

May – June 2024

- Leveraged Meta Quest 2's SDK and finger-tracking to allow real-time transcription of the user's writing to text
- Implemented a data pipeline and a CNN to translate the user's movement inputs into predicted text
- Achieved an 8.7% character error rate, as well as minimal headset energy and battery use through hosting on AWS

Speech-to-Speech Translation | *Python, PyTorch, React*

July – Aug. 2024

- Developed an end-to-end Speech-to-Speech model and app for translating real-time conversation or recordings
- Fine-tuned OpenAI Whisper and Helsinki Opus, as well as MBart + Wav2Vec, models end-to-end on English speech and translation, reaching an end-to-end ASR-BLEU score of 23.7

Threaded Twitter Server | *Go, Python, REST*

Feb. – Mar. 2024

- Built a threaded faux-Twitter server to handle RESTful API requests in Go
- Designed concurrent data structures (e.g., lock-free queues and linked lists) to enable parallel user actions
- Analyzed parallel performance and bottlenecks, collecting and visualizing latency data in Python

TECHNICAL SKILLS

Programming Languages: Python, Golang, C, C++, C#, CUDA, Java, JavaScript, Node.js

Databases/DBMS: SQL (PostgreSQL, MySQL), DynamoDB, MongoDB

Developer Tools: AWS, Git, Docker, Postman